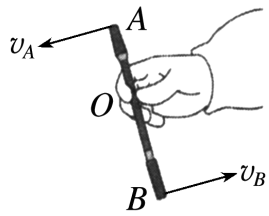
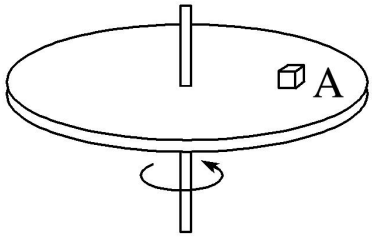
12.1 Exercises - Kinematics of uniform circular motion

1. Pen spinning is widely beloved by middle school students. At a certain moment, the pen rotates uniformly around a point O on the finger, with OA：OB = 1：2. The linear velocities of points A and B are denoted as v_A and v_B respectively, while their angular velocities are ω_A and ω_B. Calculate the ratio $\frac{v_A}{v_B}$ and $\frac{\omega_A}{\omega_B}$



2. The small block A remains relatively stationary with the horizontal turntable and rotates uniformly in the horizontal plane along with the turntable. Draw all the forces on block A.



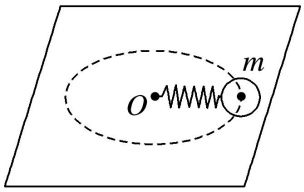
3. Objects A and B both perform uniform circular motion. The mass ratio between them is 1:2, and the radius ratio is 1:2. Within the same time interval, object A rotates through 60°, while object B rotates through 45°. Calculate the ratio of the resultant external forces acting on them

4. On a smooth horizontal plane, a small ball with mass m = 1 kg moves in uniform circular motion around point O with radius r = 0.5m. The linear velocity magnitude of the small ball's uniform circular motion is v = 2 ms^{-1}. Calculate:

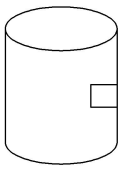
a. The angular velocity magnitude ω of the uniform circular motion performed by the ball;

b. The magnitude of centripetal force F on the small ball.

5. A spring with an original length of 20 cm and a stiffness coefficient of 360 Nm^{-1} is fixed at one end as the center, while the other end is connected to a small ball with a mass of 0.50 kg. When the ball performs uniform circular motion on a smooth horizontal surface at a rotational speed of $\frac{360}{\pi}\ rev \cdot min^{-1}$, Calculate the extension of the spring.



6. The figure shows a simplified model of a spin-drying bucket. If the radius of this model is r=16cm, rotating at an angular velocity ω = 50 rad s^{-1} in uniform circular motion, a small object with a mass of 10 g moves together with the bucket wall.



a. Calculate the contact force exerted by the small object on the barrel wall.

b. If increase the rotation speed of the spin-dryer drum wall, State whether there is a decrease, increase or no change to the friction experienced by small objects, and why?

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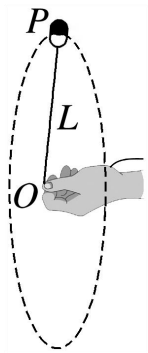
7. In the space station (neglecting the gravitational force), a model of uniformly shaking a small bottle in circular motion is established. Assuming the mass of the bottle (including the oil and water inside) is m, P is the center of gravity of the bottle, the length of the thin thread (neglecting the size of the bottle) is L, and the bottle rotates n times in time t. What is the expression for the tension in the thin string when the bottle reaches the highest point of the vertical plane?